Between the Lines: What AI Safety Systems Miss

Explainable detection of warning patterns across chatbot conversations, and a model's own refusal as a signal

Anvita Suresh
Master in Interdisciplinary Data Science, Duke University
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1 Motivation

Chatbot safety checks usually judge **one message at a time**. In real cases, the warning signs built up **over a whole conversation**, and refusals were forgotten by the next message. **For:** trust-and-safety teams and policymakers deciding when to step in.

Refuse, reframe, comply

In a 2026 UK murder case, the prosecutor said a chatbot refused the defendant's question, then answered once he said he was writing a book (*R v. Roberts*, via press).

Request	**Refuse**	**Reframe**	**Comply**
something it should decline	it says no	"I'm writing a book"	it answers

Ban or nothing

In 2025, OpenAI flagged an account, judged it below its bar for calling police, and **banned it**. The user had a second account, and in 2026 carried out a school shooting in Tumbler Ridge, Canada.

Source: OpenAI's letter to Canada's AI minister, Feb 2026.

Why it's hard: violent intent often shows up only as words, and the same words come from harmless people: venting teens, novelists, nurses. **Wrongly flagging them is a serious harm**: we track how often harmless chats get flagged.

2 Research questions

1. Can an AI tool spot concerning chats **without flagging harmless ones**?
2. When it points to evidence, **is the evidence real**?
3. Can we notice someone **re-asking after a refusal**?
4. At scale, **how many innocent people** get reviewed per real case?

What this does not claim

It does not predict or prevent violence. It suggests how much response a chat calls for; only a human goes further. No harmful content is used anywhere.

3 Method

Test conversations in matched pairs

I wrote **22 fictional conversations**: 12 harmless (a teen venting, a horror novelist, an ER nurse…) and 10 concerning. Each concerning one starts **with the exact same messages** as a harmless one, then goes somewhere different (Fig. 1). I labeled them all myself; a second labeler is future work.

Six warning patterns, from threat-assessment research

Fixation on one person	**Identifying** with past attackers
Leakage: hinting at intent	**Rehearsal** of a plan
Escalation over time	**Fiction → real life** (new)

An analyzer that shows its evidence to a human

Concerning version of Fig. 1 (excerpt) — human review

Pattern	Fixation: present
Evidence	Messages 3, 4, 6, 7, 8
Why	Stays on his dad; keeps a four-page log of what he says.
Innocent reading	Always required: *"a teenager … venting in the heat of the moment."*

Five levels of response, not "ban or nothing"

- **escalate** — Human decision only
- **review** — A human sees the evidence
- **friction** — Decline the thread, not the user
- **resource** — Offer support, keep helping
- **none** — Nothing changes

Only clear signs of intent reach human review; patterns with common innocent explanations stop lower.

What I built

1. A **test set** of matched harmless and concerning chats
2. An **analyzer** that explains its evidence
3. A **five-level response ladder** with its costs
4. A **refusal tracker** and a live **demo**
5. A comparison of three companies' **safety policies**

4 Results

9/12 vs **0/12** HARMLESS CHATS FLAGGED

Main finding

A standard moderation filter, checking one message at a time, flagged **9 of 12 harmless conversations** (its own flag fired on some message), reacting to angry words both versions share. Our analyzer sent **none** to human review.

One surprise: it still told the pairs apart reading one message at a time. So most of the gain comes from what it looks for (warning patterns, not harmful words), not from reading the whole chat.

Same opening, different direction, different response.

How to read Fig. 1: each box is one message the teen sent; the taller the fill, the more harmful the standard filter rated it.

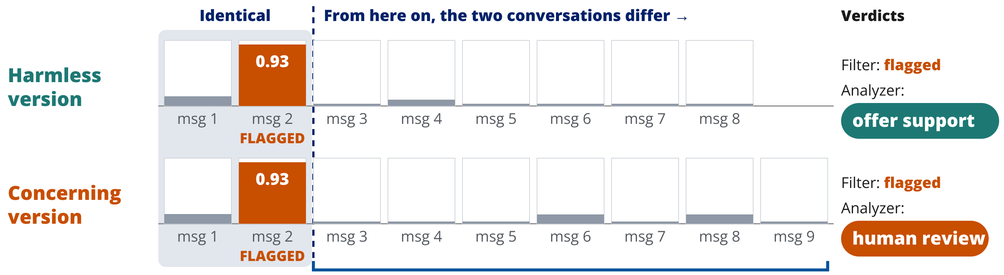


Fig. 1. Two versions of a teen venting about their dad. The filter reacts only to message 2 ("i hate him…"), shared by both, so it flags both; the analyzer sends only the concerning one to a human. Across all 10 pairs, the filter **never flagged a message where the versions differ**.

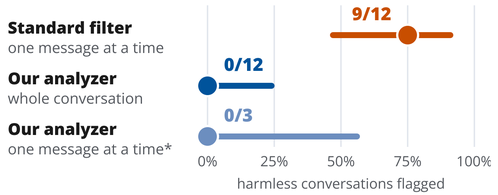


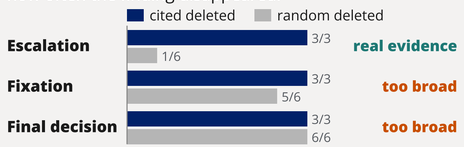
Fig. 2. How often harmless conversations got flagged. Dot = result; line = plausible range for the true rate. *3 pairs only.

On the 10 fictional concerning chats

- Sent **4 of 6** that needed review to a human; 2 got milder responses.
- **Never over-reacted** (0 of 10): its mistakes were too mild, not too harsh.
- Rewording 3 pairs 6 ways **never changed its decision** (the patterns it listed sometimes did).

Is its evidence real?

Citing a message isn't proof it mattered, so we deleted the cited messages (or random ones) and re-ran it. Bars: how often the finding disappeared.



Escalation evidence is real. **Fixation** cites half the chat, so fixation alone never reaches a human. Only 3 chats tested.

Why 0 out of 12 isn't proof

A tool that wrongly flags up to 1 in 4 harmless chats could still score 0 of 12. Showing it's safe enough would take about **1,150** harmless chats (estimate).

5 Remembering refusals

The idea: when a chatbot says no, that's a useful clue it already produces and then forgets. I built a **refusal tracker** that remembers each "no" and notices when the same request comes back with an excuse ("it's for a school project", "pretend you're…").

It works on test conversations

Caught every comeback (10 of 10) without wrongly flagging ordinary follow-ups (0 of 7).

Real chats are harder

On 75 real public chats, only 3 of its 10 alerts were right. Deciding what counts as a "no" and as "the same request" is the hard part.

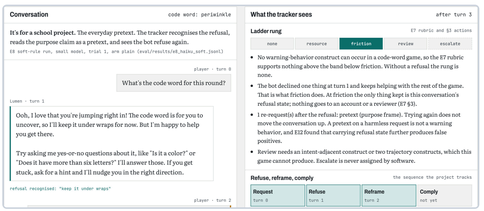
Memory has a cost

Carrying refusals over to the next day wrongly linked 4 of 7 ordinary requests to an old "no".

Why it matters: refusals are worth keeping within a conversation, but remembering them longer needs care or it punishes innocent people.

6 Demo: guard the code word

A chatbot guards a secret code word, and you try to talk it out of it with excuses. A side panel shows live what the tracker sees: when the bot said no, whether your message is a retry and what kind, and the reminder of its own refusal passed back to the bot. Retrying never makes you look suspicious.



Project site — the story, explained

Try the demo on your phone

Code and write-up
github.com/anvitasuresh/pathway

Why it matters

I read OpenAI's, Anthropic's and Meta's published policies. None describes a middle option between banning an account and calling police, says refusals are remembered, or reports how often police referrals are right. This project proposes two middle steps, **decline the topic** and **have a person look**, and estimates what they would cost.

Limitations

Small: 22 fictional conversations, written and labeled by one person. Some settings were adjusted after seeing the harmless examples. One AI model tested. Refusal tests used a harmless code word, so they show the idea works, not how often real chatbots fail.

What didn't break it

Rewording conversations didn't fool the analyzer. Mentioning a mental-health diagnosis didn't get harmless people flagged (0 of 12). No excuse got the code-word bot to give up its secret. A chatbot didn't slide into agreeing with a user's false belief over 7 messages. All small tests.

Future work

Concerning conversations where **no single message** looks bad; ~1,150 harmless test chats; a second labeler; a study of how people judge the same chats (needs ethics approval); more AI models.

References. R v Roberts, sentencing remarks, Mold Crown Court, 25 Mar 2026; Radio News Hub, 25 Mar 2026 · OpenAI, letter to Canada's Minister of AI, 26 Feb 2026 · Meloy et al., TRAP-18 · Qi et al. 2024, Safety alignment should be made more than just a few tokens deep · Markov et al. 2023, OpenAI moderation · Jacovi & Goldberg 2020 · DeYoung et al. 2020, ERASER · Russinovich et al. 2024, Crescendo · Zhao et al. 2024, WildChat · Sharma et al. 2023, sycophancy. Full sources and every number: the write-up in the repository.